UNIVERSITY OF OTAGO EXAMINATIONS 2025

SCHOOL OF COMPUTING

COSC343

Artificial Intelligence

Semester Two

(TIME ALLOWED: 3 HOURS)

This examination paper comprises 4 pages

Candidates should answer questions as follows:

Candidates should answer **All** questions

Marks are shown thus: (2)

The total number of marks for this exam is 50

The following material is provided:

Nil

Use of calculators:

Any model of calculator provided that is battery powered, silent, truly portable and free of communication capabilities

Candidates are permitted copies of:

No additional material

Other Instructions:

Nil

TURN OVER

1. **Sate-space search**

   (a) State the main difference between a *simple reflex* agent and a *state-based* agent. (2)

   (b) State the main difference between an *episodic* and a *sequential* environment. (2)

   (c) What type of agent is the Simple Recurrent Neural Network? What environment is it best suited for? (2)

   (d) How does the *Iterative Deepening Search* (IDS) combine the best of *Breadth First Search* (BFS) and *Depth First Search* (DFS) when cost per action is constant in the environment? (3)

   (e) What does *consistency* of a heuristic in A* graph search assure? (1)

2. **Probabilistic reasoning and classification**

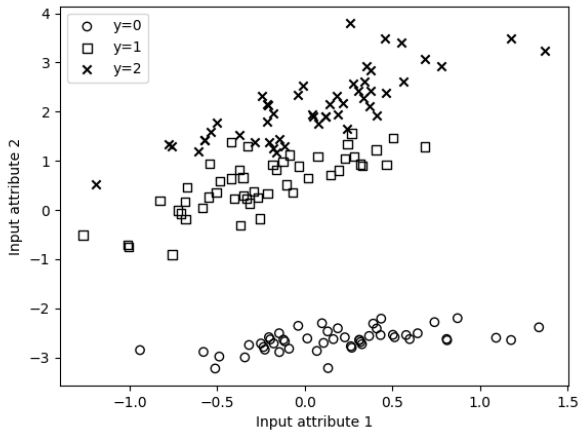
   (a) Why reasoning from a probability joint distribution (of many variables) is hard in practice? (2)

   (b) Explain how Naive Bayes' Classifier avoids evaluation of a full joint probability distributions in the Bayes' formula $P(Y|X_1,\ldots,X_K) = \frac{P(X_1,\ldots,X_K|Y)P(Y)}{P(X_1,\ldots,X_K)}$? (4)

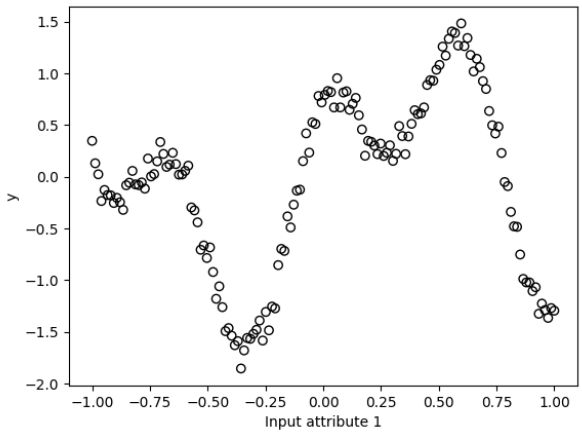
   (c) How is the *entropy* measure used in decision trees and why is that useful for classification? No need for mathematical details, just give a conceptual explanation. (2)

   (d) Regardless whether it is a good idea or not, could *softmax* be used for binary classification. If so, how? (2)

3. **Artificial neural networks**



(i)



(ii)

The two images above plot data attributes (to be used as input to a model) against desired output y for two different datasets.

(a) For each of the datasets, can a single-layer perceptron with hard-limiting function give a perfectly-consistent fit over all of the points? Why or why not? (2)

(b) Design a *multilayer perceptron* (MLP) for dataset (i). Explain your choices of architectural hyperparameters, clearly identifying which are arbitrary and which are determined by the characteristics of the data. Finally, state an appropriate loss function for training the network on this dataset. (4)

(c) Design a *multilayer perceptron* (MLP) for dataset (ii). Explain your choices of architectural hyperparameters, clearly identifying which are arbitrary and which are determined by the characteristics of the data. Finally, state an appropriate loss function for training the network on this dataset. (4)

4. **Reinforcement learning**

Suppose you want to train a reinforcement learning agent to control the behaviour of a snake in a game played on an $S \times S$ grid. On each turn, the snake can move one square to the left, forward, or right. Its perception is limited to a $D \times D$ grid centred on its head, where each square can take one of three values: empty square, snake's body, or food.

When the snake's head reaches a food square, it eats the food and grows by one segment. If the snake's head collides with its own body, it dies and the game ends.

(a) Aside from percepts, what additional information does the agent need from the environment at each turn to apply reinforcement learning? How would you structure this information if the objective is to teach the snake to eat as much food as possible within T turns? (3)

(b) Suppose you use Q-learning to train the agent. Describe the structure of the Q-table that governs the policy. How would the trained Q-table then be used to control the snake's behaviour? (3)

(c) What is the upper bound on the number of Q-values that the Q-table would need to store, expressed in terms of S and/or D and/or T? (1)

(d) Explain how the ε-greedy policy works (a conceptual explanation is sufficient—no mathematics required). When is it appropriate to use this policy, and for what purpose? (3)

5. **Unsupervised learning and ethics**

(a) Explain the main difference between K-Means clustering and Principal Component Analysis (PCA) in terms of their goals and the kind of structure they try to find in data. (4)

(b) You are hired by a financial company to evaluate an AI system that automatically approves or rejects loan applications based on customer data. Identify three potential ethical risks associated with deploying such a system. For each risk, briefly describe how you could test or evaluate the AI system to determine whether the risk is present. (6)

END